Syllabus

ECON 5021: Advanced Econometrics I

Fall 2026

Course Info	Lectures: Monday and Wednesday, 9:00 a.m. – 10:20 a.m. Location: 03-207
Instructor Info	Jonathan James Email: jjames04@calpoly.edu Office: 03-337
Office Hours	Tuesday 10:00 a.m. – 1:00 p.m.
Recommended Textbook	William H. Greene, *Econometric Analysis*, 7th or 8th Edition
Statistical Software	R/RStudio

Description

This course is a first econometrics course designed for master's students. We will cover four cornerstones of econometric analysis: statistical foundations, the linear regression model and ordinary least squares, time series econometrics, and maximum likelihood estimation. The goal is for students to develop a sound understanding of the theoretical foundations of these methods and to apply them effectively to real-world data.

Canvas Page

The course Canvas page will be the central location for course materials and announcements. Students should check Canvas regularly for lecture notes, reading lists for each lecture, practice problems for exams, information about the replication project, and datasets used for exams and in-class examples.

Expectations

You are expected to review the lecture notes and assigned readings prior to class. This includes working through the R applications in the notes as well as any proofs and derivations. Class time is intended for clarifying questions, discussion, and applications rather than first exposure to the material.

Course Schedule

Unit 1: Statistical Foundations and Computing (7 lectures)

Unit 2: Linear Regression and OLS (5 lectures)

Unit 3: Time Series Econometrics (5 lectures)

Unit 4: Maximum Likelihood and Computational Estimation (5 lectures)

Additional meetings: 2 presentation days and 4 exams

Important Dates

- 9/21: Statistical Foundations Unit Exam
- 10/12: Linear Regression Unit Exam
- 10/21: Paper Replication Proposal
- 11/2: Time Series Unit Exam
- 12/2: Maximum Likelihood Unit Exam
- 12/7 and 12/9: Replication Presentations
- 12/16: Final Exam, 7:00 a.m.–9:30 a.m.

Grading

End-of-Unit Exams (50%) There will be four end-of-unit exams, each worth 12.5% of your course grade. You will be required to do a mix of theoretical and applied work.

Final Exam (30%) The final exam will be comprehensive and will include a mix of theoretical and applied work.

Replication Proposal (5%) You will submit a proposal identifying the academic paper you plan to replicate. Additional details will be provided later in the semester.

Replication Presentation (15%) You will present your replication project to the class and provide feedback on other students' presentations. Additional details will be provided later in the semester.

Grading Scale (as a percent of total points)

		$B+ = 89.\overline{9} - 87$	$C+ = 79.\overline{9} - 77$	D	$= 66.\overline{9} - 60$
A	$= 100 - 93$	B	$= 86.\overline{9} - 83$	C	$= 76.\overline{9} - 73$	F	$= 59.\overline{9} - 0$
$A-$	$= 92.\overline{9} - 90$	$B-$	$= 82.\overline{9} - 80$	$C-$	$= 72.\overline{9} - 67$		

The grading scale above is fixed. I will not make case-by-case exceptions.

Class Policies

Make-up Exams Make-up exams will only be offered for university-approved absences.

Academic Honesty Students are expected to complete all exams honestly and independently. During exams, students may not copy another student's work, allow another student to copy their work, use outside resources, or use generative AI or code-generation tools.

Any academic dishonesty on an exam will result in an F in the course. The incident will also be reported to the Office of Student Rights & Responsibilities to determine whether further sanctions are warranted.

Special Accommodations Students who require accommodations to participate in course activities or meet course requirements are encouraged to contact the Disability Resource Center as early as possible in the semester. The Disability Resource Center can be reached at (805) 756-1395 or drc@calpoly.edu. Students with approved accommodations should notify me as early as possible so that accommodations can be implemented in a timely manner.